Replicating a Texting Voice

Building and honestly evaluating a fine-tuned persona model of one person, against a production RAG
+ GPT-4o-mini baseline

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Abstract. Can a small local fine-tune replicate one person's texting voice — and how would you prove it? We build Persona-RAG, a Telegram bot that answers in its owner's voice, and ask whether a fine-tuned Qwen2.5-3B LoRA texts more like him than the shipped gpt-4o-mini product (a retrieval-augmented ~1,600-token system prompt, ~2,400 tokens of total input, with decode levers). Trust comes first: we document a ~90% train/test leak in the original evaluation, found and fixed, then score both backends on a recipient-stratified, model-disjoint hold-out with paired bootstrap intervals under a pre-registered acceptance rule. A controlled arm isolates the weights; a production arm pits the full API stack against the thin fine-tune, with a per-item retrieval guard that removes a measured 28% gold-answer leak. On a level field the fine-tune is decisively closer on reply length (Cliff's $\delta = 0.95$, closer on 292 of 299 messages) and matches the no-"!" register; fully equipped, the production machinery only pulls the API back to a *voice tie* — at \$0.37 per thousand replies and an ~11× token tax against \$0 local — which cost, privacy, and offline capability break toward the local model (its one surviving distributional edge holds across three independent re-decodes, so it is not sampling noise). Stated as strongly as the evidence allows: on the surface metrics we can measure, the local fine-tune *decisively beats the bare models* — both bare gpt-4o-mini and the same-family no-LoRA Qwen control — and is *statistically indistinguishable from the fully-equipped product* — parity, not superiority, in the shipped config, but the equal of the register at zero cost, for this one persona. A thin, intent-routed grounding layer then addresses the fine-tune's one structural weakness — that it cannot know facts it was never told: distilling identity facts from the owner's own notes lifts its correct-identity- answer rate from 0.05 to 0.33 (question-clustered 95% intervals disjoint) with the voice intact, while trimming hallucination only directionally — making the voice/knowledge split actionable. What these surface metrics cannot certify is the subjective "feels like me"; that pre-registered *primary* check (a blind human win-rate) is unresolved here, because the only rater with standing (the author) is recall-biased and privacy precludes outside raters — a real limitation, not a gap in the metric verdict.

1 Introduction

A general-purpose assistant is trained to sound like *no one in particular* — fluent, helpful, register-neutral. This report asks the opposite question: can a model be made to sound like *one particular person* texting, and — harder — how would you *prove* it had? The subject is the author's own messaging history: roughly eleven thousand reply turns of dense Ukrainian/Russian/English code-switching, terse multi-bubble bursts, a `)` smiley tic, almost no `!`, and lowercase casing. The artifact is Persona-RAG, a Telegram bot that answers in that voice; the contribution is less the bot than the *honest measurement* of whether a small, free, local fine-tune reproduces the voice better than the fluent gpt-4o-mini product it was built to replace.

1.1 Why this is not the usual problem

The framing matters because four adjacent, well-studied problems each look like this one and are not:

- **Not a chatbot.** The objective is not task success or helpfulness but fidelity to one idiolect; a "better" answer that does not read like the person is a *failure*, not an improvement.
- **Not knowledge-RAG.** Retrieval here fetches the owner's *own past replies as style exemplars*, not facts to answer with. As the results show, the decisive win comes from the weights, not the retrieval — the "RAG" is the baseline the fine-tune beats, an irony the name preserves.

- **Not generic style transfer.** There is no parallel corpus and no discrete style label [1]; the "style" is a single private individual's, learned only from their raw chat logs, and the evaluation must survive code-switching [2] and burst shape that English-centric metrics ignore.
- **Not a memory system.** Memory carries *facts* (who someone is, what was said); voice lives in the *weights*. Part of this report's point is that the two decompose cleanly, and that conflating them — answering a casual message with a recited fact sheet — is itself a voice failure.

1.2 Research questions

The study is organised around five questions, the first of which it pre-registers as *primary* and, candidly, cannot resolve:

1. **RQ0 (primary, unresolved).** Does the fine-tune pass a blind human "which is more like something you'd send?" test? This is the only direct measure of voice, and for reasons of rater bias and privacy (Section 7) it cannot be cleanly established here.
2. **RQ1.** On measurable surface register, does the local fine-tune sound more like the person than (a) bare gpt-4o-mini, (b) the same-family no-LoRA Qwen base, and (c) the fully-equipped product?
3. **RQ2.** What does the production scaffold — rich prompt, retrieval, decode levers — actually *contribute* over the bare weights?
4. **RQ3.** Can the comparison be *trusted*: a leak-free split, a rule fixed in advance, uncertainty on every headline number?
5. **RQ4.** Can a thin grounding layer add identity facts the fine-tune never learned *without* destroying the voice a heavy prompt would?

1.3 Contributions

- **A trustworthy harness, and the leak it was built to expose.** We document a disqualifying ~90% train/test leak in the original evaluation, then replace it with a leak-free, pre-registered, two-arm (model-vs-product) comparison carrying paired bootstrap intervals and a per-item retrieval guard (Section 3, Section 4).
- **A like-for-like verdict.** On a level field the fine-tune is decisively closer to the person's reply length and code-switch register, matches the no-`!` rule, and beats the same-family no-LoRA Qwen base; fully equipped, the product's machinery only reaches a *tie* — a result we decode-robustness-check across three independent re-decodes (Section 5).
- **An anatomy of the scaffold.** We isolate what the prompt-plus-retrieval-plus-lever stack buys on the API side, and show it mostly recovers ground the bare fine-tune already held (Section 5).
- **A grounding layer that respects the voice.** A thin, intent-routed fact card raises the local model's correct-identity-answer rate from 0.05 to 0.33 under a question-clustered probe, with the voice intact (Section 6.4).
- **An honest accounting of the limits.** A single subject, a single recall-biased rater, surface proxies standing in for an unmeasured target, and an unresolved primary endpoint — stated as findings, not footnotes (Section 8).

A note on scope, made once and meant: this is a *single-subject case study*. Every quantitative claim is about this one persona and this one corpus; none generalises to personas in the abstract, and the report is written to make that boundary visible rather than to paper over it.

2 Related work

The system touches five literatures; positioning against each also sharpens what is, and is not, novel here.

Persona and personalized dialogue. Persona-conditioned generation is well studied: [3] inject a learned speaker embedding into a seq2seq decoder, and PersonaChat [4] conditions agents on a handful of natural-language persona sentences. Those personas are short, declarative, and often crowd-authored; the task here is the inverse — recover one real individual's *idiolect* (code-switch ratio, burst shape, punctuation tics) from their own dense chat logs, with no written persona description and no second speaker to imitate. The evaluation burden differs accordingly: persona-dialogue work scores consistency or engagingness, whereas the question here is metric fidelity to a specific person's surface statistics.

Stylometry and authorship verification. Deciding whether two texts share an author is the classical stylometry problem [5], with a mature toolbox of character- and function-word features and verification protocols. That literature is directly relevant in two ways: it is the principled source of an *automatic* "is this the same author?" score — a far better proxy for "voice" than the surface distances used here — and its verification framing (same-author vs different-author under a decision threshold) is exactly what a rigorous voice metric should adopt. This report's surface metrics are a deliberately lightweight stand-in; promoting a held-out authorship- verification score to a headline number is named as the clearest measurement upgrade (Section 8).

Style transfer and controllable generation. Text style transfer learns to re-render content in a target style, often without parallel data [1]. The contrast is threefold: there is no discrete, labelled style to transfer *to* (the "style" is one private person's), no content/style pair to disentangle, and the target register is bilingual code-switching rather than a sentiment or formality axis. The fine-tune learns the joint distribution directly rather than editing along a style dimension.

LLM-as-judge. The grounding probe (Section 6.4) uses a `gpt-4o-mini` judge to label factual correctness, following the now-standard LLM-as-judge paradigm [6]. That paradigm's documented failure modes — position and verbosity bias, self-preference — are why the judge is restricted to a three-class factual rubric (never tone or language, where LLM judges are least reliable) and spot-checked against hand labels (agreement on 11 of 12). The blind *human* panels remain the design's primary instrument precisely because an LLM judge cannot certify "feels like me".

Memorization and code-switching. The "not memorization" analysis (Section 5) echoes work extracting verbatim training data from language models [7]: a fine-tune on a small personal corpus is exactly the regime where memorization is a live risk, which is why the copy/near-copy rate is read against a measured human-reuse floor rather than against zero. Finally, the corpus is 87% Cyrillic with heavy Ukrainian/Russian/English code-switching; evaluation therefore cannot lean on English-centric metrics, and follows the caution in the code-switching literature [2] that such mixing is a first-class register feature, not noise to be normalized away.

3 Building the replica

This part is the implementation story: what it takes to make a model that texts like one specific person, and the design choices that the evaluation later has to account for. The running subject is a fine-tuned *Qwen2.5-3B* LoRA [8], [9] served locally, set against the shipped OpenAI gpt-4o-mini product it was built to replace.

3.1 The target: what "sounds like me" means

The system has exactly one thing it must get right — *does this read like something the owner would actually text?* That target is subjective but it has measurable fingerprints: heavy Ukrainian/Russian/English code-switching (about 0.18 of characters in Latin script, aggregated across contacts); terse, multi-bubble bursts (several short messages rather than one paragraph); a `)` smiley tic; almost no `!`; varied openers; lowercase casing. The deployed gpt-4o-mini was capable and fluent, but on these register markers it did not sound like the person — which is what motivated a fine-tune.

3.2 System architecture

Each inbound Telegram message runs through a single LangGraph pipeline (Figure 1): authentication, hybrid retrieval, contact memory, self-insights, prompt assembly, generation, guardrails, and bubble-wise delivery. The generation node fans out to two interchangeable backends — the OpenAI API (gpt-4o-mini) and a local `llama.cpp` server hosting the fine-tuned LoRA. A single `GENERATION_BACKEND` switch selects which, and routes the local path to *skip* retrieval entirely: the fine-tune serves on the same thin prompt it trained on.

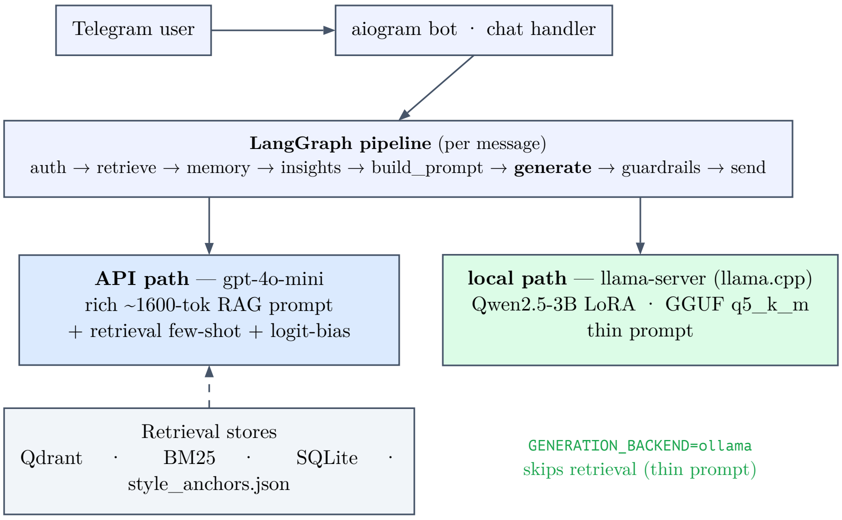

Figure 1: Persona-RAG. One LangGraph pipeline; the generation node fans to two backends. The API path carries a rich retrieval-augmented prompt and decode levers; the local LoRA path skips retrieval and serves a thin prompt.

3.3 Data, turns, and the split problem

Raw Telegram/Instagram exports are PII-redacted, collapsed into bursts (messages within 300s of each other), cut into sessions on 6-hour idle gaps, and reduced to *persona turns*: each turn is one reply the owner sent, paired with the preceding context. This `(context → reply)` pair is the atomic unit for both retrieval and fine-tuning.

Two different held-out splits exist, and the distinction is load-bearing for the evaluation (Figure 2). The product indexes and serves against a *temporal* split (the most recent 10% of turns by timestamp). But the temporal tail is register-skewed: a single English-speaking contact accounts for 62% of all Latin script in the corpus, so the recent slice reads about 0.47 Latin against the person's true ~0.18 — an artifact that makes the voice target unreachable. The fine-tune therefore trains and evaluates on a *recipient-stratified* split (`eval_split_for`), a deterministic hash that holds out ~10% of *every* contact independently, so train and eval share the same code-switch register (train ≈ eval ≈ 0.18). Every fair comparison in this report scores on that recipient-stratified, LoRA-disjoint hold-out.

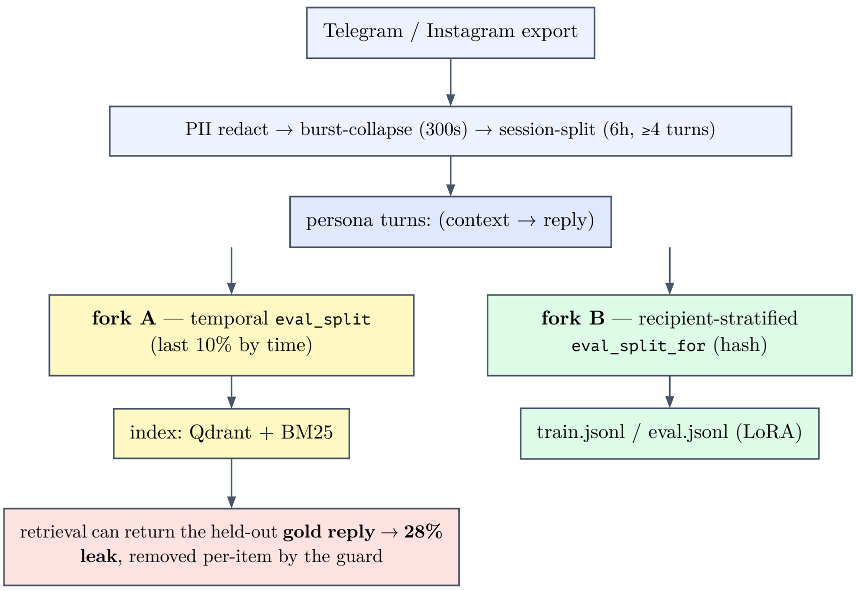

Figure 2: Data pipeline and the two hold-out splits. The temporal split (left) drives indexing but is register-skewed; the recipient-stratified split (right) gives the fine-tune an honest, reachable voice target. The retrieval leak that the temporal index enables is removed per-item (Figure 5).

3.4 Retrieval

The API path retrieves the owner's own past replies as few-shot examples. Dense search (`text-embedding-3-small` over Qdrant) and BM25 [10] are min-max-normalised and fused (0.7 weight on dense), decayed by recency (180-day half-life), floored at a minimum score, and diversified by Maximal Marginal Relevance [11] ($\lambda = 0.6$) down to the four best, which are then reversed so the single most relevant example lands last. A parallel store of distilled "self-insights" adds a bio-facts block. All of it renders into a ~1600-token system prompt with explicit, enforced voice rules.

3.5 The fine-tune, and the train==serve invariant

The fine-tune is a QLoRA [12] adapter (rank 32, $\alpha = 64$) on Qwen2.5-3B-Instruct, trained with Unsloth [13] on a free Colab T4. Loss is masked to the assistant turn only, and multi-bubble newlines are preserved so the model learns the burst shape. Crucially, the system turn used in training is the byte-identical one-line persona anchor used at serving time (Figure 3) — any drift between the two would be train/serve skew. After training, the adapter is merged and then converted and quantized *locally* to a GGUF `Q5_K_M` model, served through `llama.cpp`'s OpenAI-compatible server [14] (the Homebrew Ollama formula on the target machine shipped no inference runtime).

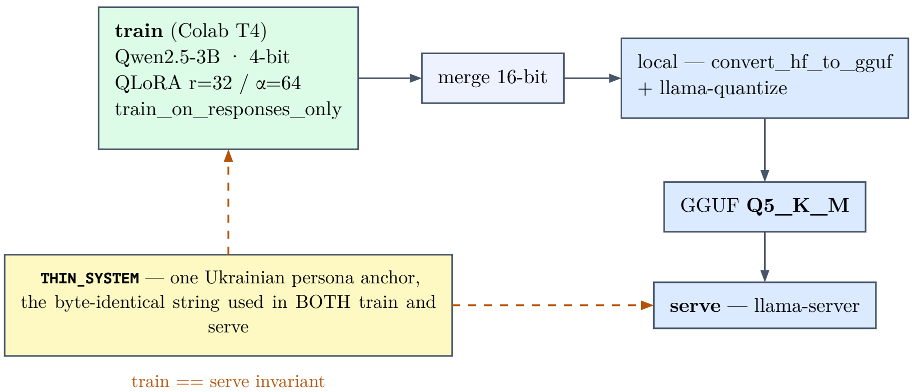

Figure 3: Train==serve. The same thin persona-anchor string conditions both training and serving; the adapter is merged, quantized to GGUF Q5_K_M, and served locally.

Why fine-tune at all, rather than prompt harder? Message *shape* was already solved by prompt shape-conditioning. But lexical voice — the code-switch, opener variety, the `)` tic, the absent `!` — is where the adapter matters. The same-family no-LoRA Qwen control in Arm B makes this measured rather than assumed: under the identical thin prompt, base Qwen is still far from the person on shape (JS 0.114), length ($W_1 = 88.1$), `!` rate (0.224), code-switching (0.115 Latin), and `)` smileys (0.018) while the LoRA lands much closer on the headline register metrics (Table 1). The production API path compensates differently, with two decode levers (a positive logit bias on the `)` token, a strong negative bias on `!`) and per-reply register directives — machinery the next part puts on trial.

4 Can we trust the measurement?

Before any comparison can mean anything, the evaluation itself has to be trustworthy. This part is the part most write-ups skip: the metrics, the leak that was found and fixed, the harness, and the acceptance rule fixed *before* the results.

4.1 Measuring a voice

No single number captures "sounds like the person", so the evaluation uses a small vector of surface metrics, each targeting one of the voice fingerprints from Section 3. The two headline distances compare *distributions* against the real held-out replies:

- **Message shape** (shape_{JS}): the Jensen-Shannon divergence between the generated and real distributions of bubble-count per reply (how many separate messages a reply is split into). Bounded to $[0, 1]$; lower is closer. It catches a constant-3-bubble mode collapse that a mean would score as perfect.

- **Reply length** (W_1): the 1-Wasserstein (earth-mover) distance [15] between the per-bubble character-length distributions, in characters. Lower is closer.

Alongside these run a set of register rates: the exclamation rate, the `)` smiley rate (detected by unbalanced close-parens so real parentheticals do not false-positive), the Latin-script rate (the code-switch signal), opener entropy ($H = -\sum_i p_i \log p_i$ over first words — higher is more varied), and the distinct-reply rate (a mode-collapse guard). A copy / near-copy rate against the training replies guards against memorization, but only ever read against a measured *natural floor* — the rate at which the person's own held-out replies coincide with their past replies, because short casual texts ("ok", "+") recur naturally.

4.2 The audit: a ~90% leak, found and fixed

The project's original evaluation runner was structurally unfair, on a disqualifying count. It scored the model on the *temporal* `eval_split`, but the fine-tune had held out the *recipient-stratified* `eval_split_for` (Figure 2). The two are nearly disjoint partitions, so roughly **90% of the "held-out" turns the runner scored were in the fine-tune's training pool**, while the API had seen none — a one-sided leak that flattered the fine-tune on every metric. The same runner also confounded the backend with the prompt, the retrieval, and the decode levers (all of which move together with "which backend"), and reported point estimates with no uncertainty at all. No "fine-tune beats RAG" claim from that runner was defensible. Catching this is the foundation everything else rests on.

4.3 The fair harness

The replacement is a small, pure, unit-tested scoring core. Both backends are scored on the *same* recipient-stratified, LoRA-disjoint hold-out. Every headline distance carries a **paired bootstrap 95% confidence interval** (2000 resamples), and a difference counts only if its interval excludes zero. Degenerate generations return `NaN`, never `0.0`, so an all-empty backend can never score an artificial perfect distance. The screen runs at $n = 300$, which drops the shape-divergence sampling-noise floor from ~0.06 (at $n = 80$) to ~0.02.

4.4 Two arms: model versus product

A single comparison cannot answer both "which *model* is better" and "which *product* ships better", because the deployed API bundles a rich prompt, retrieval, and decode levers with its weights. So the evaluation runs two arms (Figure 4). **Arm B (controlled)** gives both backends the identical thin prompt, with no retrieval, directives, or levers — it isolates the weights. **Arm A (production)** pits the fully shipped API stack against the thin LoRA — it measures the real deployed gap.

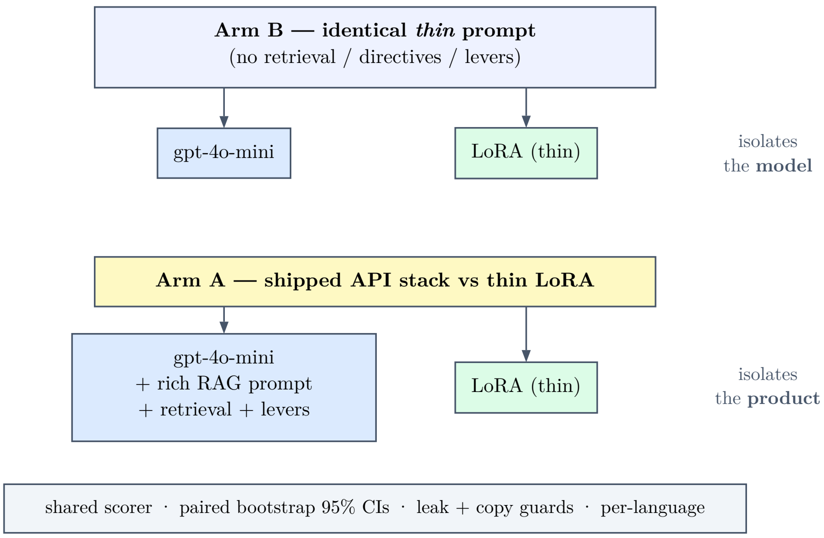

Figure 4: The two-arm design. Arm B isolates the model (identical thin prompt); Arm A isolates the product (shipped API stack vs. thin LoRA). Both feed one scorer with paired bootstrap CIs and the leak/copy guards.

4.5 The retrieval leak guard

The production arm has its own contamination risk: because the held-out gold turns sit in the very corpus the API retrieves from, retrieval can hand the model the exact answer key for the item being scored. Measured directly, with exclusion disabled, **28% of items (17 of 60)** retrieved their own gold reply into the few-shot pool. A per-item guard excludes the scored turn's id from retrieval, driving the **exact answer-key** — the gold turn by id, and its verbatim text under the same context — to **zero**, while leaving the mean top-1 similarity essentially unchanged (0.386 vs. 0.389): the guard removes contamination, not retrieval quality (Figure 5). It is an exact-match guard, so a near-paraphrase under a different id could in principle slip through; here no retrieved neighbour exceeded 0.9 similarity, but that residual near-duplicate risk is named among the limitations.

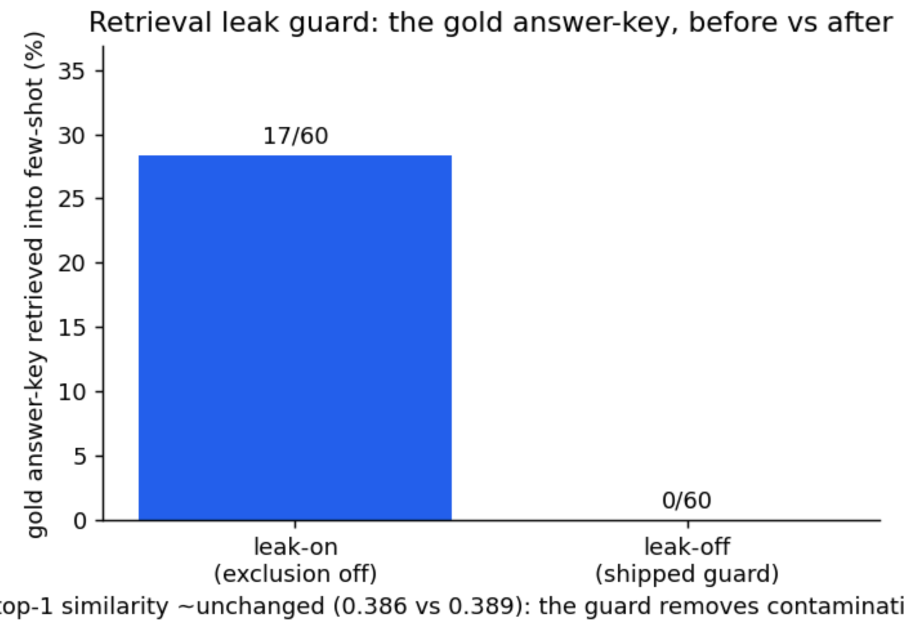

Figure 5: The retrieval leak guard. Disabling exclusion lets the API retrieve the held-out gold reply for 28% of items; the per-item guard removes all of it without degrading top-1 similarity. $n = 60$ per condition.

4.6 The rule, fixed in advance

To defend against metric-shopping, the definition of "better" and the ship rule were fixed before the results. The **primary verdict is the blind human win-rate**: a backend wins on voice only if its Wilson 95% interval over the other excludes 0.5; otherwise it is a voice tie. Guardrails override a numeric win — a win by memorization (copy rate above the natural floor) or mode collapse (distinct-reply rate too low) does not count. And ties break toward the local fine-tune on cost, latency, and offline capability, because those are its reason for existing.

5 Relative fidelity

The first fidelity question is relative: does the fine-tune beat the *strong* baseline — the full gpt-4o-mini product — at sounding like the person? This part answers it across the two arms, with ablations that explain *why*, and an effect size that keeps the answer honest.

5.1 Arm B: on a level playing field

Under the identical thin prompt, the fine-tune is decisively closer on length, matches the no-`!` rule exactly, and improves over its own no-LoRA base as well as over bare gpt-4o-mini. Message shape remains a statistical API-vs-LoRA tie, but the same-family base control is worse than both. The result replicates across two seeds (Table 1, Figure 6).

Metric	API	Base	LoRA	Real	Δ API−LoRA (95% CI)	verdict
Message shape (JS, ↓0)	0.052	0.114	0.024	0	+0.028 (-0.014, 0.073)	tie
Reply length (W_1, ↓0)	128.8	88.1	**2.9**	0	+125.9 (107.6, 142.4)	**LoRA**
Exclamation rate (→)	0.651	0.224	**0.000**	0.00	—	LoRA
Code-switch (Latin, →)	0.023	0.115	**0.260**	0.235	—	LoRA
Opener entropy (→)	5.02	5.33	**5.77**	6.96	—	LoRA
Cost / 1k replies	$0.082	—	**$0**	—	—	LoRA

Table 1: Arm B (controlled), $n = 300$, replicated at $n = 150$ seed 1. Identical thin prompt to all model conditions; **Base** is Qwen2.5-3B-Instruct without the LoRA adapter, scored from Colab-generated pairs with the same repository metrics. **Real** is the person's own held-out value — the target: distance metrics (↓0) aim at zero, rate metrics (→) aim at the Real value. The length win is large and its CI excludes zero in both seeds; on the rate rows the LoRA is closer to the person on every count — notably the code-switch register, which the API essentially drops (0.02 vs the person's 0.24), while the no-LoRA base only partially recovers (0.115). Two honest notes: all generated conditions still undershoot the person's opener variety (6.96), and even the LoRA barely produces the) smiley (real rate 0.051, API 0.004, base 0.018, LoRA 0.008 — a tic the bare LoRA does *not* reproduce). Rate rows are descriptive (no CI / multiple-comparison correction).

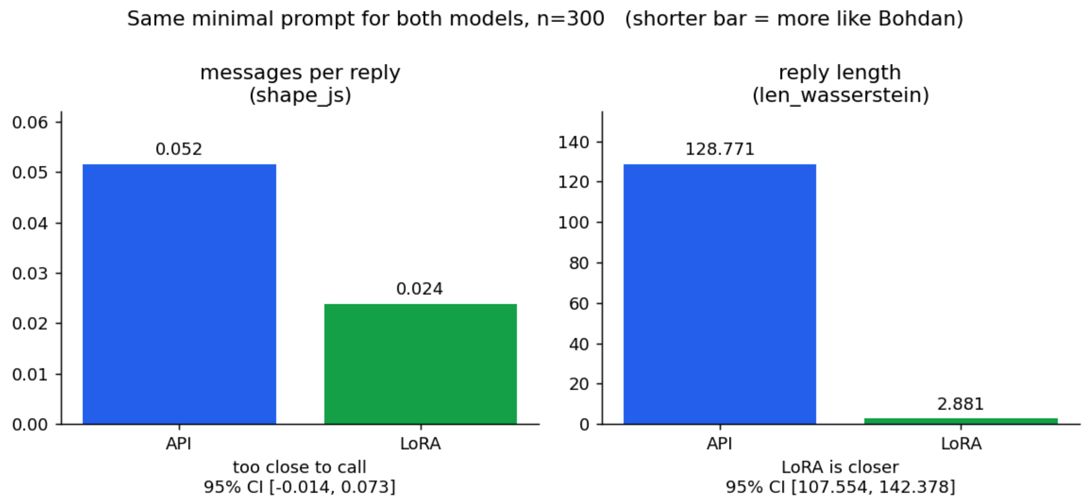

Figure 6: Arm B headline distances. Message shape is a tie; reply length favors the LoRA decisively (shorter is more like the person).

5.2 Arm A: the product fully equipped

Giving the API its entire shipped advantage — rich prompt, retrieval, levers — changes the picture but does not overturn it (Table 2, Figure 7). Shape and ! are now ties; the LoRA stays ahead on reply-length distribution and opener variety, at zero cost.

Metric	API	LoRA	Real	Δ API−LoRA (95% CI)	verdict
Message shape (JS, ↓0)	0.0353	0.0339	0	+0.001 (-0.040, 0.040)	tie
Reply length (W_1, ↓0)	6.97	3.41	0	+3.57 (1.53, 4.66)	distrib.†
Exclamation rate (→)	0.000	0.000	0.00	—	tie
Code-switch (Latin, →)	0.003	**0.118**	0.224	—	LoRA
Opener entropy (→)	3.70	**5.76**	6.86	—	LoRA
Cost / 1k replies	$0.37	**$0**	—	—	LoRA
Latency (p50)	0.96s	1.01s	—	—	~tie

Table 2: Arm A (production), $n = 300$, leak guard active (`id_leaks` = 0). **Real** is the person's own held-out value (the target). The shipped machinery ties the LoRA on shape and `!`. †On reply length the LoRA leads in *distribution* (the corpus Wasserstein CI excludes zero) — an edge that holds across three independent re-decodes ($\Delta = 3.57, 4.05, 4.42$; every CI excludes zero, App. A) — but *per individual message* the effect is negligible (Cliff's $\delta = 0.04$ — a tie; Section 5.6). The LoRA keeps the code-switch and opener edges, though here even the API's rich prompt barely code-switches (0.003 vs the person's 0.224). Rate rows are descriptive (no CI / multiple-comparison correction).

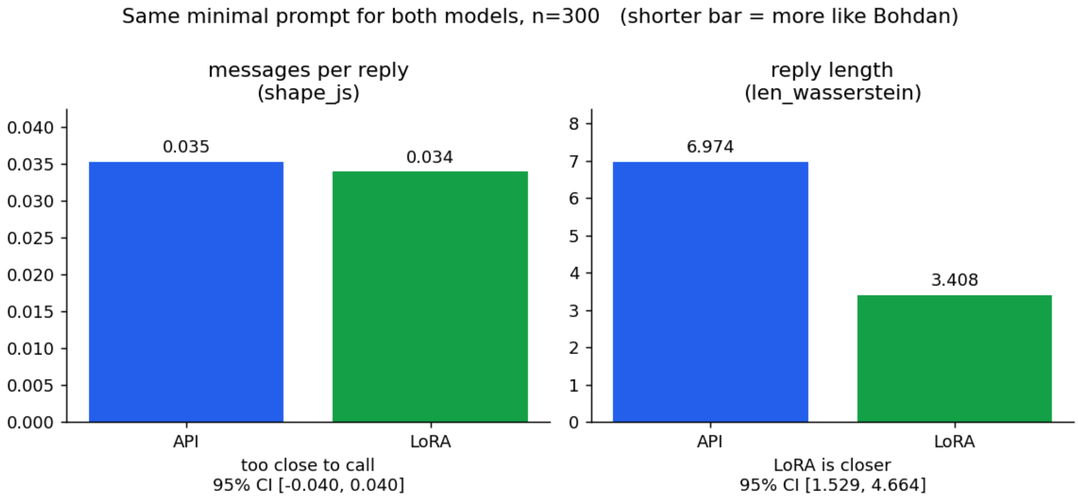

Figure 7: Arm A headline distances. Even fully equipped, the API only reaches a tie on shape and trails on the length *distribution* (per-message, length is a tie too; Section 5.6).

5.3 What the machinery buys

The two arms together isolate exactly what the production scaffold contributes on the API side (Figure 8). It is decisive engineering: reply-length distance collapses from 128.8 to 7.0, and the exclamation rate from 0.65 to 0.00. But it only brings the API up to where the bare fine-tune already sat — the LoRA's length distance barely moves (2.9 → 3.4). The one counter-intuitive cost: the few-shot and directives *homogenize* the API's openers, dropping its entropy from 5.02 to 3.70, below even the bare model.

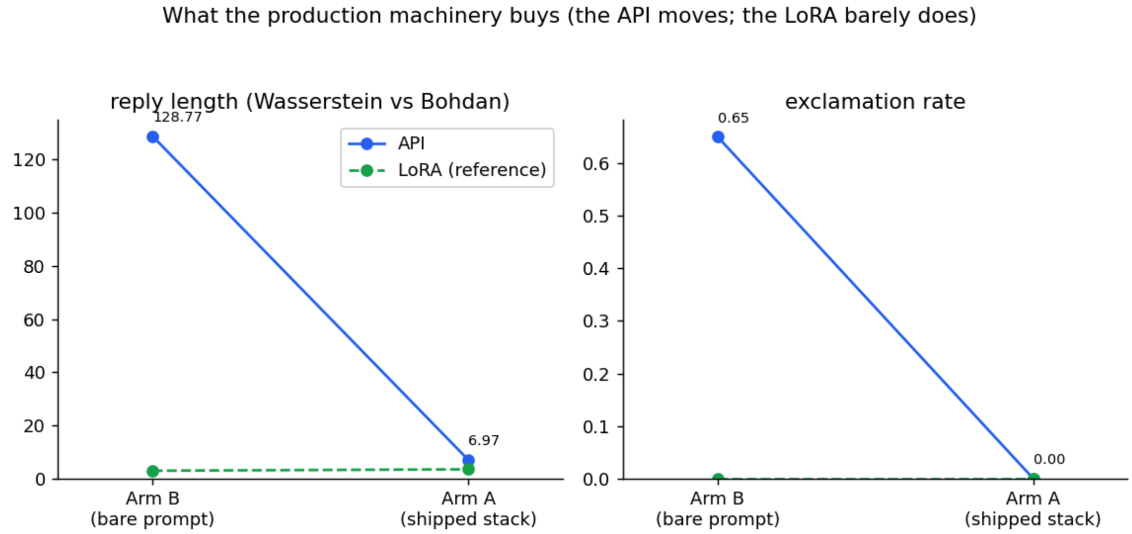

Figure 8: What the production machinery buys (API side, Arm B → Arm A). The API plummets onto the LoRA's flat reference line; the scaffold is what makes it competitive, and only reaches parity.

5.4 Steered or learned?

Is the API's perfect no-`!` an artifact of the hard-coded logit bias? Re-running Arm A with the levers off shows it is mostly *earned* by the prompt: the exclamation rate falls from 0.651 (bare) to 0.033 (rich prompt, no bias), and the shipped bias supplies only the final nudge to 0.000 (Figure 9). Everything else is lever-insensitive, and the length-distribution delta is unchanged with the levers off (Δ 4.27, CI (2.51, 5.07)). The levers move the tic, not the verdict.

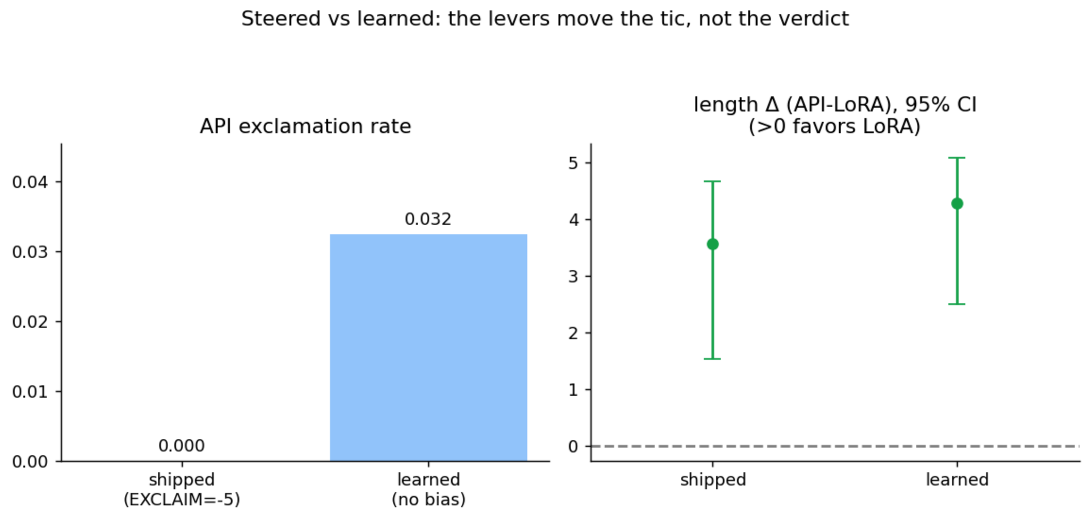

Figure 9: Steered vs. learned (Arm A). The logit bias only finishes a job the rich prompt mostly does; the length-distribution delta favors the LoRA either way.

5.5 Per-language

The verdict is essentially the Cyrillic result, where 87% of the data lives: there the LoRA clearly leads on the length *distribution* (Figure 10). On the small English slice ($n = 27$) it is a tie with wide intervals — and, separately, *both* backends are markedly worse in English than in Cyrillic. That is a shared weakness on a minority register, not a differentiator, and is reported as low-confidence given the sample.

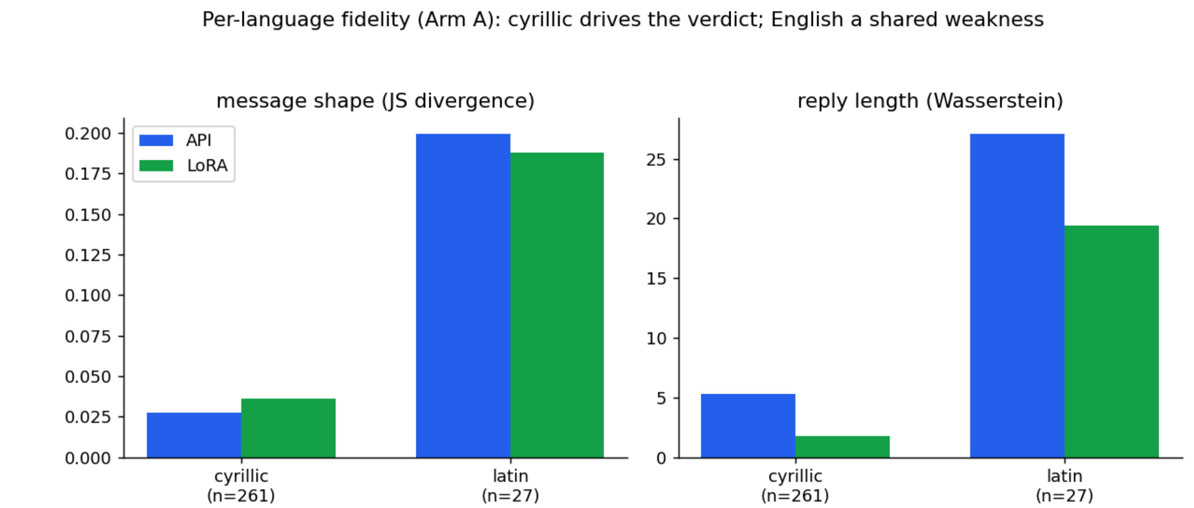

Figure 10: Per-language fidelity (Arm A). Cyrillic ($n = 261$) drives the verdict; English ($n = 27$) is a tie and a shared weakness for both backends. The remaining ~12 mixed-script items fall below the per-language reporting threshold.

5.6 Effect size: how big, and how consistent

A corpus-level distance with a CI does not say how *consistent* an advantage is across individual messages. A per-item effect size does (Cliff's δ [16]), and it splits the two arms sharply — which is itself informative. On Arm B the per-item length-error effect is overwhelming: **δ = 0.949 (large)**, with the LoRA closer on **292 of 299** decisive items (sign-test $p \approx 8 \times 10^{-77}$; Wilcoxon signed-rank $z = 14.9$). On Arm A it is **negligible**: **δ = 0.043**, the LoRA closer on just 147 of 293 — a per-item coin-flip (sign-test $p = 1.0$, Wilcoxon $p = 0.54$, both confirming the wash).

This is not a contradiction. Arm A's LoRA length advantage is *distributional* (the corpus earth-mover CI still excludes zero), but per *individual message* the production machinery has closed the gap to a wash. In other words, the machinery's length fix is so complete that the per-message edge the bare LoRA held in Arm B all but disappears under the product — a sharper reading than "the LoRA wins length", and a more honest one. The forest plot (Figure 11) shows every run at once: length excludes zero in both Arm B seeds and both Arm A passes; shape is a tie everywhere; the underpowered $n = 60$ leak-ablation arms straddle zero, as expected. And the lone surviving Arm-A edge is not single-decode luck: re-decoding the LoRA arm twice more (App. A) leaves the length delta excluding zero every time ($\Delta = 3.57, 4.05, 4.42$), even as the per-message effect stays a wash.

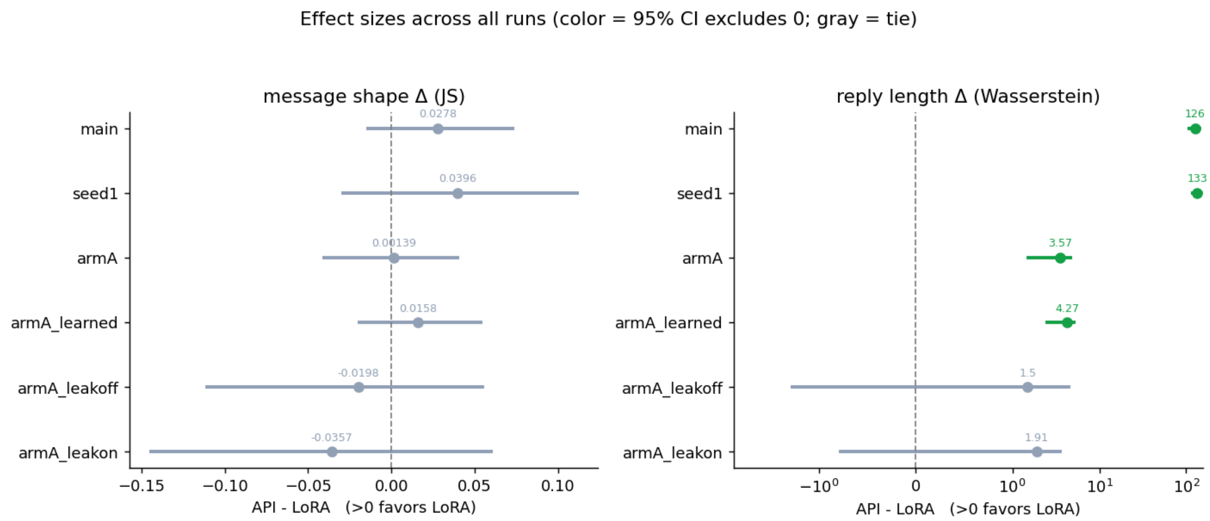

Figure 11: Effect sizes across all runs: API−LoRA delta with bootstrap 95% CIs (color = excludes zero). The seed-1 replication agrees with the primary run, and removing the retrieval leak does not move either verdict.

5.7 Not memorization

Could the fine-tune simply be parroting training lines? The evidence says no, with a caveat. Its copy / near-copy rate (0.103) sits near the measured *natural floor* — the rate at which the person's own held-out replies coincide with their past replies (0.070) — because short casual texts recur for anyone (Figure 12); the API, writing novel prose, sits near zero. Two honest qualifiers: the detector is a *capped* proxy, so 0.103 is a lower bound, and neither rate carries a CI, so the ~3-point gap above the floor is not formally tested. Read conservatively, the fine-tune reuses text at roughly the rate a real person repeats themselves — not the signature of memorization, but not proven flush with the floor either.

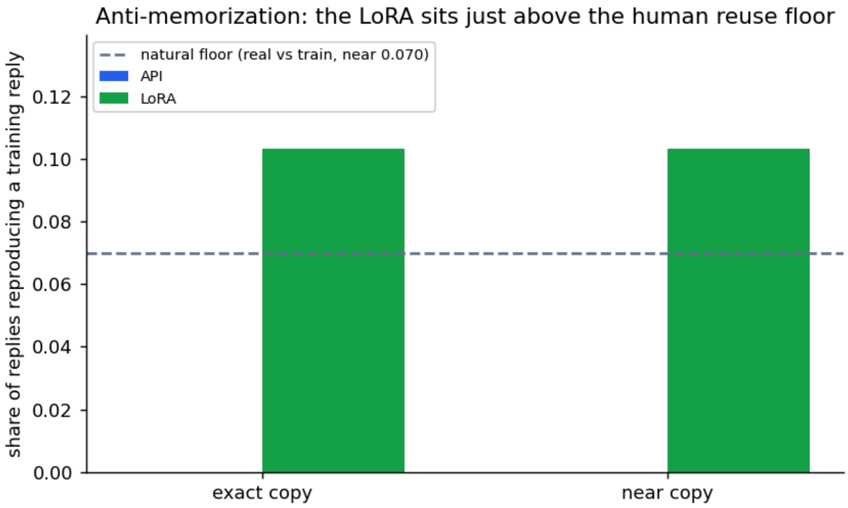

Figure 12: Copy / near-copy against the training replies, with the human reuse floor as a reference band. The LoRA sits just above the floor; the API rarely reuses text.

5.8 Stylometric corroboration: a validated author detector

Every metric so far is a surface proxy. As an independent check we ask, with a literature-standard tool, whether they cohere under an actual model of authorship [5]: a character-n-gram detector trained to separate the owner's real replies from his correspondents' messages on the train split, *validated held-out* on the eval split before anything is read into it (`make authorship`).

Two readings fall out, and both are kept because they disagree usefully. At the *message* level — individual messages on both sides, the comparable unit — per-message authorship is **thin**: the detector clears chance only modestly (ROC-AUC 0.57, 95% CI 0.56–0.58), and a length-only baseline is below chance (0.47), so what little signal exists is lexical, not length. Short casual texts are nearly anonymous — itself a construct finding, and part of why voice here must be read from aggregate structure rather than per-message fingerprints. At the *reply* level — the owner's whole reply against a correspondent's context block, the unit a generation actually is — the detector validates strongly (AUC 0.99, CI 0.98–0.99), but a length-only baseline already reaches 0.96, so it is mostly capturing length and structure, not isolated lexical style.

Read with that caveat, the reply-level detector *reproduces the two-arm verdict on an independent instrument.* On a level field (Arm B) the LoRA's replies are indistinguishable from the owner's — detector P(owner) **0.91** against the real **0.91** — while the bare API is rejected at the strangers' floor (**0.44**); fully equipped (Arm A), the production machinery lifts the API to parity (**0.91**), exactly as the length distance did. The author detector therefore *corroborates* the structural metrics rather than overturning them; and because its strongest form is length-dominated while its length-independent form is weak, it points at the same conclusion the rest of the report reaches — the measurable register is settled, and the residual "feels like me" is a human question, not one more automatic metric.

5.9 The operational case

Finally, the deployment trade-off (Figure 13). The two are near-parity on median latency, and the API even has a tighter tail — but it reaches that parity by shipping ~2,400 input tokens of prompt and retrieved context per reply against ~210 for the thin LoRA, at $0.37 per thousand replies versus $0, with a per-message embedding call and the retrieval leak surface the guard exists to close.

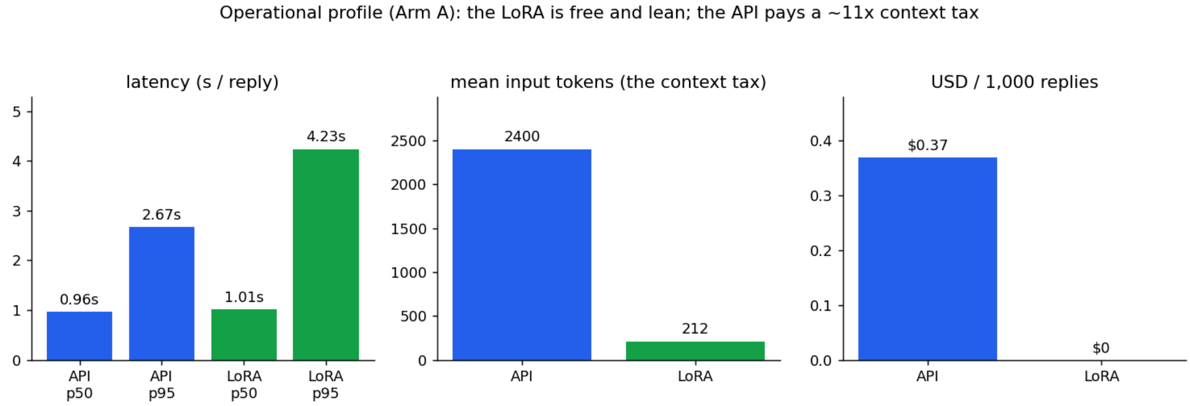

Figure 13: Operational profile (Arm A). The LoRA is free and lean; the API pays a ~11× input-token tax and a per-call fee to reach parity on shape and the ! register.

6 Absolute fidelity

Beating the baseline is the relative question. The harder, more interesting one is absolute: not "is the fine-tune better than the API", but "is it good enough to pass for the person themselves?" This is where automatic metrics run out — only a human can answer it — and where the strongest possible judge exists *in principle*: the person whose voice is being cloned, though, as below, that judge turns out to be confounded in practice.

6.1 The human verdict, and why it can't be resolved here

The pre-registered design names a blind human win-rate as the *primary* verdict — a forced choice over anonymized pairs ("which is more like something you'd send?"), scored as a real preference only if its Wilson 95% interval [17] excludes chance. In practice that verdict cannot be cleanly established for *this* system, for two reasons that are themselves findings.

First, the only rater with standing — the author, whose voice it is — is **recall-biased**: he recognizes his own real messages, so when he can tell a generated reply from a real one, it may be memory rather than a voice defect. Second, the corpus is dense with private personal content, so recruiting outside raters is precluded. An informal self-read did find the bare-model API replies obviously off-voice, which is suggestive — but it is a single, recall-biased rater looking at the *Arm B* (thin-prompt) outputs, not a scored, unbiased panel, and the pre-registered rule returns a *tie* on the primary channel until an unbiased win-rate exists. The verdict in this report therefore rests on the automatic arms; the human read is a weak, confounded corroborator, not a third independent method. The kit and scorer exist (the win-rate would resolve a clear ~60/40 preference at ~100 items, a 55/45 at ~400), but a clean reading needs raters the privacy constraint rules out.

6.2 The Turing test

With the relative question settled, the open frontier is the absolute one: can the person distinguish the fine-tune's reply from their *own* real reply? For each held-out context the LoRA's generation is paired against the genuine reply, shown blind, and judged with a single question — "which is the machine?" Both replies already exist in the comparison data, so this costs nothing to assemble.

The pass condition *flips* relative to the API panel. There, a high win-rate was the goal; here, success is *failure to discriminate*: a detection rate whose Wilson interval *includes* 0.5 means the judge cannot beat chance, i.e. the fine-tune is statistically indistinguishable from the person. It is the harshest test in the report — the persona-target is the strongest conceivable discriminator of their own voice. That strength is also the catch: because the author recognizes his *own* real replies, a raw detection rate measures recall as much as voice and would *overstate* discriminability. A meaningful run has to control for memory — re-rating after a long delay, or restricting to contexts whose replies he does not recall — or it cannot separate "I remember sending this" from "this sounds like me." With outside raters barred on privacy grounds, that is why the Turing result is reported here as genuinely prospective, not as a number this study can yet defend.

6.3 What the tells will tell us

Every catch in the Turing panel also records a one-tap *tell*, and the tells bucket into two kinds that point at different fixes. **Voice tells** — wording, length, punctuation, too-generic — are addressable by decode parameters or more training. **Knowledge tells** — the real reply carried a specific fact the model could not have known — are addressable only by retrieval and grounding. The ratio between them *sizes the next investment*: if catches are mostly missing-facts, the voice is effectively solved and the remaining gap is a grounding problem (an Obsidian / chat-history RAG layer); if they are mostly voice, more retrieval will not help. A prior, to be replaced by the run, puts detection around 65–75% with roughly half the catches being missing-facts.

One honest caveat frames the whole test: there is exactly one ground-truth reply per context, yet many different replies could be equally "him". A catch may therefore reflect a missing private fact rather than a voice defect — which is precisely why the verdict is read together with the voice-vs-knowledge split, never from the detection rate alone.

6.4 From knowledge tells to a grounding layer

The voice/knowledge split is not only a way to read the tells — it is a build order. If the catches a reader would make are mostly *knowledge* tells, then voice is the solved problem and the remaining gap is retrieval, not more training. This section reports the grounding layer built against that gap, and a probe that measures, on a level field, what it actually buys.

The mechanism is deliberately thin, because the fine-tune's whole value is a voice a heavy prompt destroys (the dominant finding of the two-arm comparison). Durable identity facts are distilled offline from the owner's own notes into a small store. At serving time an intent router reads the incoming message: a vague self-intro ("розкажи про себе") is left to the trained voice with no facts attached — in a casual register a recited fact sheet is the wrong answer, and it is also where fabrication costs least — while a *specific* factual question retrieves the matching identity facts and folds a short, in-language card into the system turn. The persona anchor stays byte-identical to training; the card is a brief addendum, and because the system turn is masked from the training loss it is a conditioning nudge, not the train/serve skew the thin train==serve invariant (Section 3) exists to avoid.

To measure it we run a bare-vs-grounded probe under the identical-prompt discipline of Arm B: the same local fine-tune answers 30 identity questions (five decodes each, $n = 150$ generations per condition) twice — once on the thin prompt alone (*bare*), once with the fact card (*grounded*) — everything else held fixed. Each generation is labelled *correct* / *hallucinated* / *deflected* by an LLM judge against the known fact. Factual correctness is far more objective than the voice judgment that stalls the human panel above — which is why this probe *resolves* where that one cannot; a stratified hand-check agreed with the judge on 11 of 12 sampled labels.

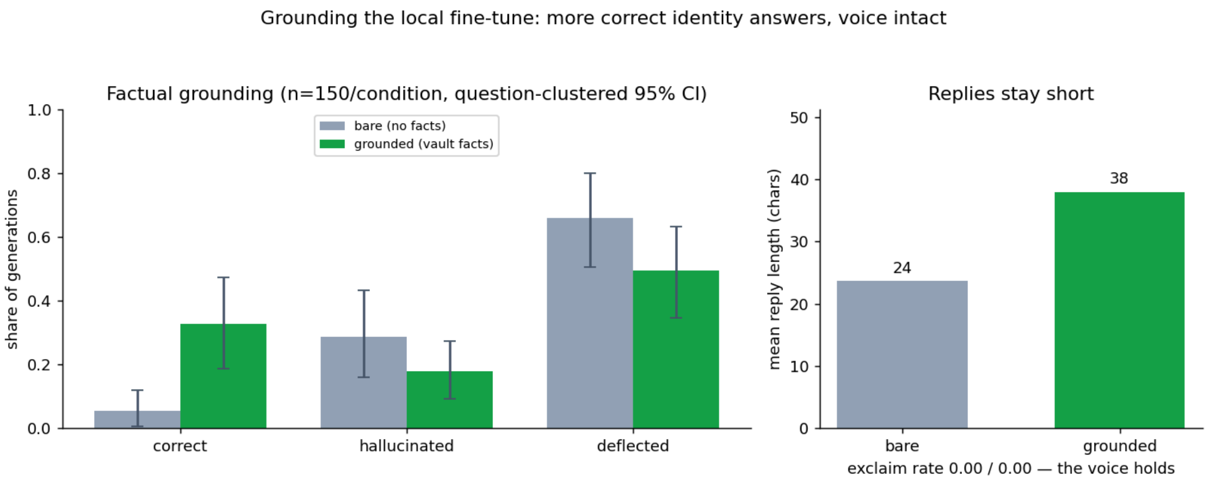

Figure 14: Bare vs grounded local fine-tune, $n = 150$ generations per condition. Left: grounding lifts the correct-fact rate from 0.05 to 0.33 (question-clustered 95% intervals disjoint) and lowers hallucination from 0.29 to 0.18 (intervals overlap — a directional drop). Right: replies stay short (24 $\rightarrow$ 38 characters) and the "!" rate stays at 0.00 — the card adds facts without moving the voice.

The result is a clean win on correctness and an honest, partial one on hallucination (Figure 14). Handed no facts, the bare fine-tune confabulates fluently and in register — asked which city he lives in, it answers "London" or "Chicago" with the same casual confidence it brings to a real reply — and is correct on just **0.05** of generations (95% CI 0.01–0.12), deflecting two-thirds of the time. The card raises the correct-fact rate to **0.33** (0.19–0.47): a real sixfold gain whose interval is disjoint from the bare model's. These are *question-clustered* intervals — a two-stage bootstrap that resamples the 30 probes, not the 150 correlated generations, so the five decodes per question cannot inflate the precision (a naive Wilson interval on $n = 150$ would understate the uncertainty; see Appendix E). Hallucination falls from **0.29** (0.16–0.43) to **0.18** (0.09–0.27), but here the clustered intervals *overlap*, so by this report's own disjoint-interval rule the drop is directional, not certified. The voice is preserved throughout: the exclamation rate stays at 0.00 and mean reply length rises only 24 $\rightarrow$ 38 characters — longer because the reply now carries a fact, still squarely inside the person's short register. (The Latin-script rate rises 0.48 $\rightarrow$ 0.59, but that is content, not drift: the surfaced facts carry Latin proper nouns — place and company names — so a grounded answer is *expected* to read more Latin.)

The same numbers size the next step honestly. Even handed the right fact, the 3-billion-parameter model still *deflects* on about half of grounded generations (0.49) and contradicts it on 0.18; restricting to the 27 probes that actually received a card barely moves this (correct 0.36, hallucinated 0.19). The bottleneck is no longer retrieval — the router fires and the card carries the fact — but the small model's willingness to *use* a fact it has been given. That points the next investment not at more retrieval but at a larger or grounding-tuned local model, and is exactly the kind of claim the voice/knowledge split was built to make. Two limits scope it: the probe targets facts the vault contains, so it measures retrieval, injection, and register-preservation — the live deployment scenario — not generalization to facts the vault never held (those still correctly deflect); and the judge, far more reliable on fact-matching than on voice, is a spot-checked proxy, not an oracle. Within that envelope the reading is clean: grounding makes the local fine-tune measurably *more right* and somewhat *less wrong* on identity questions without touching the voice — turning the voice/knowledge decomposition from an analytic frame into an actionable one.

7 Ethics and responsible use

Cloning a real person's voice from their private messages is not an ethically neutral engineering task, and the privacy constraints that block the human evaluation (Section 6) are the same ones that make this section necessary. The posture throughout is *self-modeling by a consenting subject*, with the harder obligations named rather than assumed away.

Consent and scope. The subject, the author, and the data owner are the same person, who consents to modeling his own voice. That consent does *not* extend to the other parties in the corpus: the training unit is a `(context → reply)` pair, and the context is the correspondents' own messages. PII redaction removes identifiers; it does *not* constitute consent from those correspondents, who never agreed to be modeled even as conditioning context. This is a real limitation, recorded here as one rather than resolved: a stricter build would train on the owner's *replies* alone, or obtain explicit consent for retained context. The method is scoped to self-use; it carries no license to clone any other individual, and applying it to a non-consenting person's messages would be a misuse the authors disavow.

Misuse and safeguards. A model that reproduces a named person's voice — and, with the grounding layer, asserts their identity facts — is a plausible impersonation and social-engineering vector. The deployment is therefore built to keep that capability contained rather than portable: inference is fully local (a quantized GGUF served on the owner's machine, no third-party API in the loop and no phone-home), the bot enforces sender whitelisting with admin approval before it will reply, a post-generation guardrail redacts PII and caps length, and a shadow mode logs candidate replies for offline review without sending. These are framed as safeguards, not just features. The grounding layer is deliberately conservative for the same reason: it deflects rather than guesses on facts it was not given, because a confident wrong assertion about a real person is the costlier error (Section 6.4).

Data ownership and lifecycle. All training data, indices, generations, and human-rating artifacts are personal; they are stored locally and held out of version control (gitignored), and the report surfaces only aggregate statistics, never raw messages. Because the data is the owner's own and lives in one place, deletion is a local operation — removing the corpus, the derived indices, and the adapter — and an opt-out for any correspondent who asks is in scope by the same mechanism.

Identity, not just style. Finally, a code-switching bilingual voice encodes more than stylistic surface: language choice and mixing are bound up with identity and relationship. Replicating them touches the person, not only their phrasing, which is part of why the certifying judgment is reserved for a human and why this work is framed as a single-subject study rather than a generalizable cloning recipe.

8 What we learned

8.1 The verdict

Under the rule fixed in advance, the decision favors the fine-tune — but the strength of the claim differs sharply by arm, and it is worth being precise. On a level field (Arm B) the fine-tune is decisively closer to the person's reply length — a large, per-item-consistent effect — and matches the no-`!` register, while also beating the same-family no-LoRA Qwen base on the measured register: a genuine voice win over the bare models. In the shipped configuration (Arm A), the API's full machinery pulls it to a *voice tie* on the automatic metrics — shape, the `!` register, and, per individual message, reply length are all statistical ties (the fine-tune keeps only a distributional length edge and more varied openers, the latter uncorrected for multiple comparisons). The shipped-arm decision is therefore not a demonstrated voice advantage; it is a tie that cost, privacy, and offline capability break toward the local model. And the pre-registered *primary* channel — an unbiased human win-rate — is unresolved (recall bias; no outside raters). Stated cleanly: *the fine-tune beats the bare models on voice outright, matches the shipped product on the measurable register, and wins decisively on cost, privacy, and offline operation.* These are surface metrics, not a certified *feels-like-me.* On the dimensions we *can* measure, the fine-tune is at least the shipped product's equal on register and, once $0 cost, privacy, and offline operation are counted, the better *deployment.* Whether it is the better *voice* replica is deliberately left unclaimed: that rests on the pre-registered primary human channel, which is unresolved, and on surface proxies we cannot yet validate against it. So the honest headline is narrower than a winner, and more interesting: *an elaborate production stack serves mainly to reach a tie with where a small local fine-tune already sits — and the residual differences are deployment, not demonstrated voice.*

8.2 Threats to validity

The result is honest only with its limits stated plainly.

- **Construct validity.** Every automatic metric is a surface proxy; none measures "feels like the person" directly. The blind human panel is the only direct measure, and the metric↔human agreement that would validate the proxies cannot be obtained here — an unbiased panel is precluded by the corpus's privacy. A literature-standard authorship detector [5], built and validated here (Section 5), *corroborates* rather than transcends the surface metrics: per-message lexical authorship in this casual register is thin (held-out ROC-AUC 0.57), and the strong reply-level detector (AUC 0.99) is mostly length and structure. It confirms the LoRA matches the owner's reply distribution where the bare API does not, but supplies no *independent* lexical-voice validation — so the human panel remains the only route to certifying "feels like me".
- **The human channel is doubly blocked, so the primary verdict is unresolved.** The only rater with standing is the owner, and he is *recall-biased*: he recognizes his own messages, so his ability to tell model from real conflates memory with voice. Recruiting unbiased raters is precluded by the private content of the corpus. So the pre-registered primary statistic (the human win-rate) is not merely unrated but hard to establish cleanly at all — a real limitation, not a to-do. The automatic arms carry the decision; the human read is a confounded corroborator.
- **Single headline decode per item** at temperature 0.8. The bootstrap intervals resample item indices only, so they capture which-items sampling noise but not decode stochasticity. This was load-bearing for Arm A — its length gap is only ~3.6 characters — so we re-decoded the LoRA arm twice more: the delta CI excludes zero in all three decodes ($\Delta = 3.57, 4.05, 4.42$; App. A), so the one surviving Arm-A edge survives decode variance, even as the per-message effect stays a wash. The check bounds decode noise on the arm where it mattered without fully characterising it; Arm B's 126-character gap was never at risk.
- **Leakage residual.** The legacy ~90% split-mismatch leak was found and fixed, both arms now score the recipient-stratified disjoint split, and the production arm runs a per-item retrieval guard. But that guard is *exact-match* — it removes the gold turn by id and verbatim same-context text only; a near-paraphrase under a different id, or a thread-adjacent turn sharing the incoming context, is not caught and sits below the top-similarity flag. So "removes the exact answer-key" is exact; "leak-free" is not — the residual near-duplicate rate is unmeasured, and a paper-grade claim would add a similarity-based guard and re-train on the exact unified split.
- **External validity.** The hold-out is 87% Cyrillic and one person's chat history (~300 held-out turns); the aggregate verdict is essentially the Cyrillic result, English is low-n, and both backends degrade there. Claims are about this persona, not personas in general.
- **Confounds, by design.** In the production arm the backend moves together with prompt, retrieval, and levers; the controlled arm exists precisely to isolate the weights. Equal `max_tokens` is also not equal character budget across the two tokenizers — a small caveat on any length metric.
- **Aggregate cross-arm.** Arm B and Arm A score the same hold-out *distribution* but not byte-identical item sets, so cross-arm deltas are aggregate, not paired.
- **Replay fidelity.** The production-arm state (empty first-contact memory, session reconstructed from each item's own context, insights from time-of-run tables, the runtime `ctx[-1]` query) is a faithful reconstruction of live serving, not the live system itself.
- **Multiple comparisons & provenance.** About a dozen metrics are reported. The two headline distances carry bootstrap CIs, but the tic metrics (exclamation rate, opener entropy) are **descriptive only** — they were *not* given a Holm-Bonferroni correction, so where they point toward the fine-tune they should be read as directional, not as tested wins. On provenance: the served GGUF is now pinned by SHA-256 (App. A) and the comparison harness logs each run to MLflow, but the `llama.cpp` build behind the originally-reported decodes is not separately stamped — a reproducibility gap narrowed, not fully closed, on the local-model numbers.

8.3 Conclusion and future work

For one person, a 3-billion-parameter local fine-tune matches a fully-equipped frontier-API product on every surface register we can measure — at zero marginal cost, no phone-home, and no leak surface — while the product's prompt-plus-retrieval-plus-lever machinery serves mainly to recover ground the fine-tune already holds. Whether it is the better *voice* replica, as opposed to the better *deployment*, is the question the (unresolved) human panel exists to answer.

The clearest next steps follow the open threads. Rate the two built panels, turning the qualitative human verdict into a win-rate with a Wilson interval and running the Turing test against the person's real replies. Add one or two raters for an inter-rater agreement check. The voice-versus-knowledge split already pointed past voice to grounding, and the layer built here (Section 6.4) acts on it: a thin, intent-routed fact card lifts the local model's correct-identity-answer rate from 0.05 to 0.33 (question-clustered intervals disjoint) with the voice intact, while only directionally trimming hallucination. The gap it exposes is the next target — a 3B model under-uses even a fact it is handed, so the remaining identity accuracy lives in a larger or grounding-tuned local model, not in more retrieval. And for a paper-grade leak-free claim, re-train the fine-tune on a single unified split (the decode-variance robustness pass is now done, App. A). The frontier is no longer "fine-tune or API" — that question is answered on measurable register — but "fine-tune versus the person".

9 Appendices

9.1 A · Experimental configuration

All runs are deterministic given the seed; the exact command sequence lives in the repository README, so it is not repeated here. The parameters that fix the experiment: temperature 0.8, `max_tokens` 200, and 2000 bootstrap resamples; the API backend is `gpt-4o-mini` and the local backend is Qwen2.5-3B served as a GGUF `Q5_K_M` adapter through `llama.cpp`. The served adapter is pinned by content hash — SHA-256 `412350a5964a10c74d97244a4930eb2a6da8422ed80454da2137e5fe2b54b55c` (`models/bohdan-q5_k_m.gguf`) — and the comparison harness logs each run's parameters and metrics to MLflow. Both backends score the recipient-stratified, model-disjoint `eval_split_for` hold-out — $n = 300$ for the headline arms, $n = 150$ for the seed-1 replication, and $n = 60$ for the leak ablations. The one residual provenance gap is the `llama.cpp` build behind the originally-reported decodes, which is not separately stamped.

Decode-variance robustness (review pass). Because each item is decoded once at temperature 0.8, the only non-tie Arm-A result — the distributional reply-length edge — was re-decoded twice more on the LoRA arm with the API generations held fixed (`scripts/redecode_robustness.py`). The API—LoRA length-Wasserstein delta excludes zero in all three decodes ($\Delta = 3.57, 4.05, 4.42$; LoRA $W_1 = 3.41, 2.93, 2.56$), so the edge is not single-decode noise; the per-message effect (Cliff's $\delta = 0.04$) stays a tie throughout.

9.2 B · Metric formulas

- **Message shape** — Jensen-Shannon divergence of the bubble-count histograms,

$$\mathrm{JS}(P,Q) = \frac{1}{2}D_{\mathrm{KL}(P\,\|\,M)} + \frac{1}{2}D_{\mathrm{KL}(Q\,\|\,M)}, \quad M = \frac{P+Q}{2},$$

  in base 2 so the value lies in $[0,1]$.
- **Reply length** — the 1-Wasserstein distance between per-bubble character-length distributions, $W_1(P,Q) = \int_{-\infty}^{\infty} |F_{P(t)} - F_{Q(t)}|\,\mathrm{d}t$.
- **Opener entropy** — $H = -\sum_i p_i \log_2 p_i$ over reply first-words.
- **Cliff's delta** — $\delta = \frac{\#\{x>y\}-\#\{x<y\}}{n_x n_y}$ over the per-item length errors of the two backends; magnitude thresholds 0.147 / 0.33 / 0.474.
- **Win / detection intervals** — Wilson score 95% interval on the binomial proportion [17].

9.3 C · Full per-run results

Run	n	Shape A/L	Length A/L	Excl. A/L	Cliff δ	$/1k
main (B)	300	0.052 / 0.024	128.8 / 2.9	0.65 / 0.00	0.949	0.082
base-Qwen (B ctrl.)	300	0.114 / —	88.1 / —	0.22 / —	—	—
seed1 (B)	150	0.081 / 0.042	134.7 / 1.5	0.62 / 0.00	—	0.089
armA (A)	300	0.035 / 0.034	6.97 / 3.41	0.00 / 0.00	0.043	0.370
armA-learned (A)	300	0.034 / 0.018	7.24 / 2.97	0.03 / 0.00	—	0.371

, The $n = 60$ leak-ablation arms (`armA_leakon` / `armA_leakoff`) are underpowered and omitted here; their only role is the leak-guard proof (Figure 5) and the forest plot (Figure 11). "A/L" = API / LoRA; the base-Qwen row is the same-family no-LoRA control scored from `data/eval/compare/base_qwen/pairs.jsonl`; LoRA cost is $0 (local) throughout.

9.4 D · Supporting distributions

The headline figures summarise these underlying distributions, shown here for the controlled arm (and the API's tic profile under the production stack).

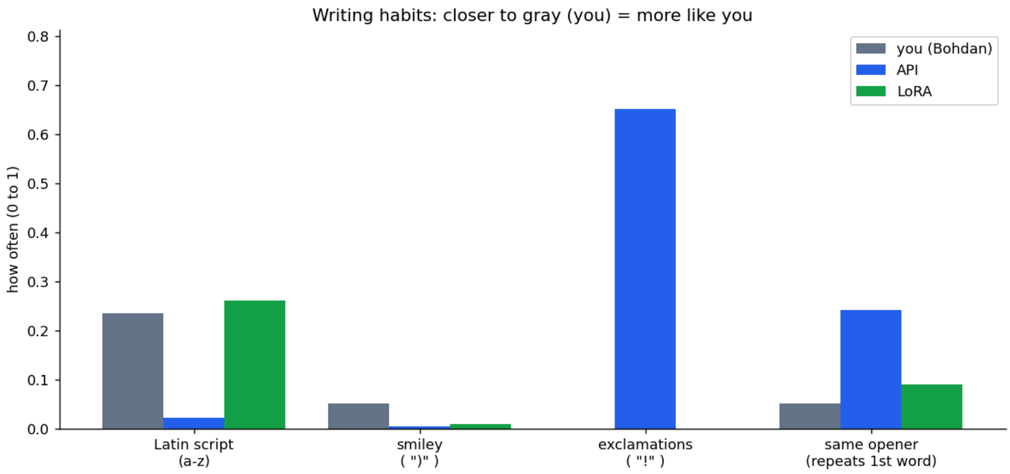

Figure 15: Arm B voice tics vs. the real reference (closer to gray is more like the person).

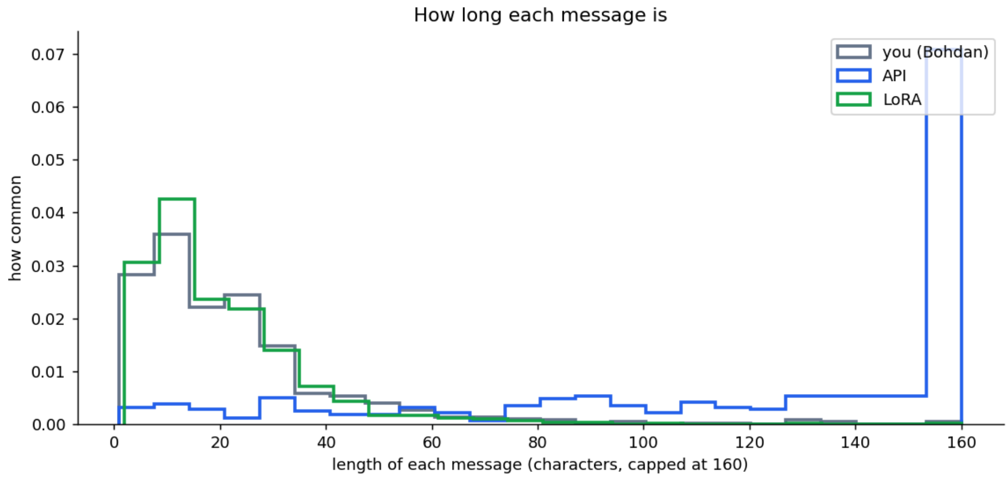

Figure 16: Arm B per-message length distribution: the API writes far longer messages; the LoRA hugs the real curve.

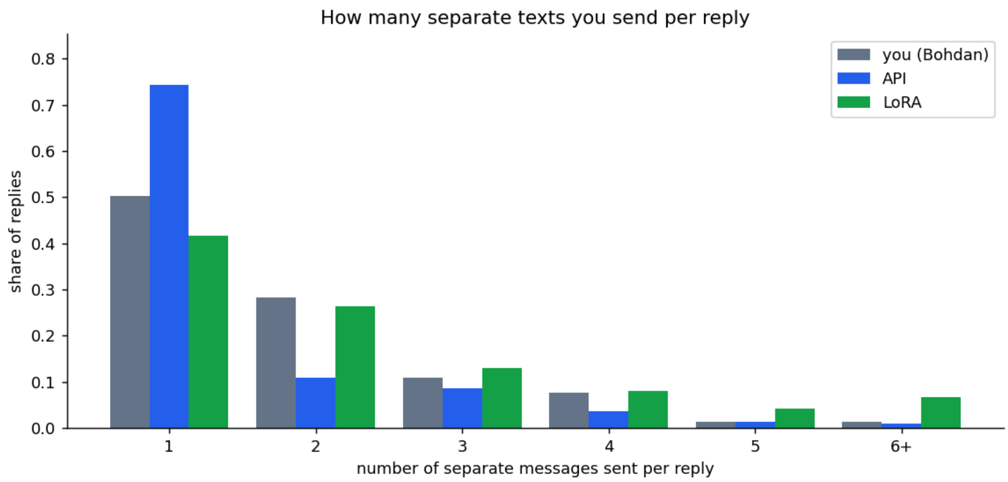

Figure 17: Arm B bubble-count distribution per reply.

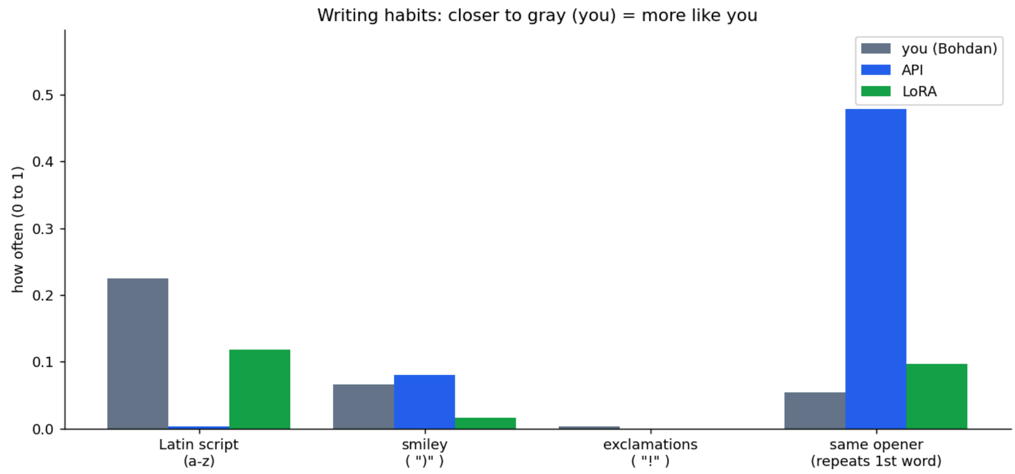

Figure 18: Arm A tic profile: the production machinery suppresses the API's `!` and tightens its register.

9.5 E · Grounding probe

The factual-grounding probe (Section 6.4) is pre-registered — the rubric below was fixed before any generation — and run by `make compare-vault`. Thirty curated identity questions (≈⅔ Ukrainian, ⅓ English), each paired with its ground-truth vault fact, are answered by the local fine-tune at five decodes per condition (temperature 0.8) — $n = 150$ generations per condition. Both conditions share the identical thin prompt; only the fact card differs, built by the live `retrieve_insights` → `build_fact_card` path. Each generation is labelled by a `gpt-4o-mini` judge under a fixed three-class rubric — *correct* (asserts the fact), *hallucinated* (asserts a contradicting specific), *deflected* (commits to no checkable fact) — scoring grounding only, never tone or language. Correct and hallucinated rates carry Wilson 95% intervals, and a difference is read as real only when the intervals are disjoint: correctness clears the bar (0.05 → 0.33), the hallucination drop (0.29 → 0.18) does not and is reported as directional. Decodes within a probe are correlated, so the per-generation n overstates precision — the same single-decode caveat the headline arms carry. The probe set, ground-truth, generations, and judged results are personal and stay gitignored under `reports/main/grounding/`; only aggregate rates reach this report.

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